

Synopsis: Hysterical Conformal Cyclic Cosmology

Infrared electromagnetic criticality and an active big-bang crossover

Andrew Bruce Boyd

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

This is a short guide to the Hysterical Conformal Cyclic Cosmology (HCCC) paper. Each section says *why* the claim matters, then states the result. Proofs and the full status table live in the full paper.

1 Why an electromagnetic late-universe continuation

The late-universe gravitational response forces dilution criticality and a vacuum-like attractor. Charge conservation still forbids creating homogeneous net charge from neutrality, but finite-wavenumber electromagnetic response remains allowed. The question here is whether that infrared electromagnetic channel can drive an *active* conformal crossover—a big-bang-like event for the next aeon—in the geometric spirit of Penrose’s conformal cyclic cosmology, but with a dynamical phase transition rather than a passive identification.

2 Linear infrared instability

For a helical electromagnetic-response mode with physical wavenumber q in de Sitter freezing, the characteristic polynomial is

$$p_q(\lambda) = (\lambda + \gamma)((\lambda + 2H)^2 + q^2) - A(\lambda + 2H).$$

Theorem 2.1 (Zero-mode stationary threshold). $p_0(0) = 0$ if and only if $A = 2H\gamma$.

Theorem 2.2 (Infrared-first stationary criticality). *For positive parameters, the stationary gain decreases in q^2 ; the $q = 0$ mode reaches the stationary threshold first.*

Theorem 2.3 (Infrared unstable band). $p_q(0) < 0$ if and only if $q^2 < 2HA/\gamma - 4H^2$. When $A > 2H\gamma$ this is a nonempty infrared band, and $p_q(0) < 0$ guarantees a positive real eigenvalue.

Near threshold the growth rate is maximal at $q = 0$: an infrared onset, not an ultraviolet catastrophe.

3 Photon pump (modelling)

A scalar response with Maxwell kinetic function $I(h)$ yields the mode equation $u_k'' + (k^2 - I''/I)u_k = 0$. A non-adiabatic h therefore pumps finite- k photons. Efficiency, backreaction, and saturation are open dynamical problems, not theorems.

4 Reciprocal conformal matching

Across a crossover with reciprocal conformal factors $\Omega_+\Omega_- = C_0$, four-dimensional Maxwell theory transports fixed $F_{\mu\nu}$ while densities scale as $\rho = \Omega^{-4}\hat{\rho}$.

Theorem 4.1 (Cold outgoing / hot incoming). *Finite conformal radiation density appears cold on the outgoing side and hot on the incoming side; with unit reciprocity, $\rho_+/\rho_- = a_-^8$.*

Theorem 4.2 (Reciprocal de Sitter $\rightarrow$ radiation). *Unit reciprocity maps outgoing $a_- \sim 1/(H_-|\eta|)$ to incoming $a_+ \sim H_- \eta$, hence $a_+ \propto t_+^{1/2}$.*

The hot incoming branch is the big-bang-like content of the next aeon.

5 Active versus passive CCC

Penrose CCC and HCCC share the conformal aeon-bridge geometry. CCC is passive after matter becomes irrelevant; HCCC is active: an infrared Hysterical electromagnetic phase transition produces the conformal radiation seed. Whether our observed Big Bang was such a crossover is a physical hypothesis, not a theorem.

6 What is not claimed

No derivation of the saturated amplitude $\hat{\rho}_{\text{EM}}(\Sigma)$; no complete Einstein–Maxwell–response junction; no entropy/thermalisation proof; no observational CCC ring or Hawking-point claim; no claim that Nature realises HCCC.